

CM2 cont.

$$\min_{A \in \mathbb{R}^{N \times M}, N \geq M} \min_{x \in \mathbb{R}^M} \|Ax - b\|^2$$

$$A = \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix} \quad b = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}$$

$$\nabla f(x) = A^T A x - A^T b$$

$$\nabla^2 f(x) = A^T A \quad : \text{Matrice de Gram.}$$

$$(A^T A)_{ij} = \langle a_i, a_j \rangle, \quad A = (a_1 \dots a_n)$$

$$A^T A x = A^T b$$

$A^T A$ défini positive:

$$x^T A^T A x = (Ax)^T Ax = \|Ax\|^2 \geq 0.$$

$$\text{Si } x^T A^T A x = 0 \Rightarrow Ax = 0 \Rightarrow x = 0 \Rightarrow A^T A \text{ SDP}$$

$$\boxed{\text{Supp } A \text{ inj} \Leftrightarrow A \text{ rang } M}$$

• coercif? $\lim_{\|x\| \rightarrow \infty} f(x) = +\infty$

$$\lambda \neq 0$$

$$f(\lambda x) = \lambda^2 x^T A^T A x - \lambda (Ax)^T b + \|b\|^2$$

$$\exists \beta > 0 \quad \text{t.q.}$$

$$\|Ax\| \geq \beta \|x\| \Rightarrow f(x) \geq \beta^2 \|x\|^2 - C \|x\| + \|b\|^2 \Rightarrow f \text{ coercif,}$$

"Inj quantitative"

$\|A \cdot\|$ est une norme

$$\|Ax\| \geq \beta \|x\|$$

$$\text{Soit } \varphi(x) = \|Ax\|$$

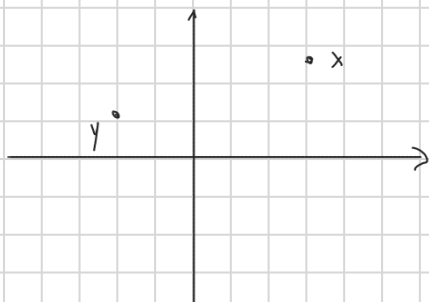
$$\min_{\|x\|=1} \varphi(x)$$

C^0 sur cpt

$$\exists x^* \text{ t.q. } \varphi(x^*) = \min_{\|x\|=1} \varphi(x) \Rightarrow \forall x \text{ t.q. } \|x\|=1 \Rightarrow \|Ax\| \geq \beta$$

$$\Rightarrow \forall x \quad \|Ax\| \geq \beta \|x\|$$

Soit x, y deux minimiseurs de f .



$$\text{Soit } \varphi(t) = f(x + t(y - x))$$

$$\varphi(0) = \min = \varphi(1)$$

$$\varphi'(t) = (y - x)^T \nabla f(x + t(y - x))$$

$$\varphi(t+h) = \dots$$

$$\varphi''(t) = (y - x)^T \mathcal{H}_f(y - x)$$

$$\mathcal{H} = \nabla^2 f(x + t(y - x)) = A^T A$$

$$\varphi''(t) \geq \beta \|y - x\|^2$$